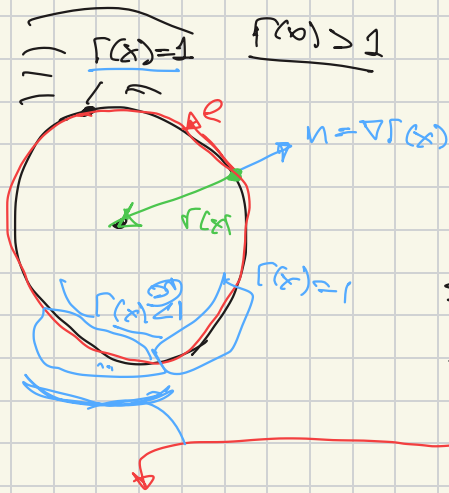




Normal-based Modulation



$\Gamma(x)$ c'
 $n = \nabla \Gamma(x) \checkmark$
 $e = \nabla \Gamma(x)^\perp \checkmark$

$$E_n = [n \ e]^T$$

$$D = \begin{pmatrix} \lambda_n & 0 \\ 0 & \lambda_e \end{pmatrix}$$

$$M = E D E^{-1}$$

$$\Downarrow$$

$$M_n = E_n D E_n^T$$

$$\left[\lambda_n = 1 - \frac{1}{\Gamma(x)}, \lambda_e = 1 + \frac{1}{\Gamma(x)} \right]$$

$$E_r = [r \ e]^T$$

$$M_r = E_r D E_r^{-1}$$

reference-based modulation

Exercise #1

$$\hat{x} = \underbrace{E D E^{-1}}_{\Gamma(x)} f(x)$$

far from obstacle $\Gamma(x) \rightarrow \infty$

$$\lambda_n(x) = 1 - \frac{1}{\Gamma(x)}$$

$$\Gamma(x) \rightarrow \infty$$

$$\frac{1}{\Gamma(x)} \rightarrow 0$$

$$\lambda_n(x) \rightarrow 1$$

$$\lambda_e(x) = 1 + \frac{1}{\Gamma(x)}$$

$$\Gamma(x) \rightarrow \infty$$

$$\frac{1}{\Gamma(x)} \rightarrow 0$$

$$\lambda_e(x) \rightarrow 1$$

Closer to the obstacle $\Gamma(x) = 1$

$$\lambda_n(x) = 1 - \frac{1}{1} = 0$$

$$\lambda_e(x) = 1 + \frac{1}{1} = 2$$

Exercise #2

$f(x) \xrightarrow{x \rightarrow \infty} 1/x \rightarrow 0$

Show there is a fixed point $\dot{x} = M(x)f(x) = 0$ @ $x \neq x^*$

$$M(x) = E(x) D(x) E(x)^T$$

$$E(x) = [n \ e], D = \text{diag}[\lambda_n \ \lambda_e]$$

$$\dot{x} = E(x) D(x) E(x)^T f(x)$$

$$= [n \ e] \begin{bmatrix} \lambda_n & 0 \\ 0 & \lambda_e \end{bmatrix} \begin{bmatrix} n^T \\ e^T \end{bmatrix} f(x)$$

$$= [n \ e] \begin{bmatrix} \lambda_n n^T f(x) \\ \lambda_e e^T f(x) \end{bmatrix}$$

$$= \underbrace{n \lambda_n n^T f(x)}_{\lambda_n = 1 - \frac{1}{f(x)}} + \underbrace{e \lambda_e e^T f(x)}_{\lambda_e = 1 + \frac{1}{f(x)}} = 0 \quad ?$$

$$\lambda_n = 0, \Gamma(x) = 1$$

on boundary $\Gamma(x) = 1$
 $\Rightarrow \lambda_n = 0$

$$\Gamma(x) = 1$$

$$\lambda_e = 2$$

$$2 e (e^T f(x)) = 0$$

$$n \lambda_n n^T f(x) = 0$$

"Independability"

$$e \perp f(x)$$

$$e^T f(x) = 0$$

$$n(x)^T f(x) = \pm 1$$



Collinearity
 solve
 ff

Exercise #3

Show that $M_r = E_r D E_r^{-1}$ the updated DS

$\dot{x} = M_r f(x)$ will have fixed point $\dot{x} = 0 \neq \underline{x}^*$

$$E_r = \begin{bmatrix} r & e \end{bmatrix}$$

$$\lambda_r = 1 - \frac{1}{F(x)} \\ \lambda_e = 1 + \frac{1}{F(x)}$$

$$E = \begin{bmatrix} n & e \end{bmatrix}$$

$$\dot{x} = \begin{bmatrix} r & e \end{bmatrix} \begin{pmatrix} \lambda_r & 0 \\ 0 & \lambda_e \end{pmatrix} E_r^{-1} f(x)$$

$$E_r = E \Sigma \Rightarrow \Sigma = E^T E_r$$

$$\Sigma = \begin{bmatrix} n & e \end{bmatrix}^T \begin{bmatrix} r & e \end{bmatrix} = \begin{bmatrix} n^T r & 0 \\ e^T r & 1 \end{bmatrix}$$

$$\Sigma^{-1} = \begin{bmatrix} \frac{1}{n^T r} & 0 \\ -\frac{e^T r}{n^T r} & 1 \end{bmatrix}$$

$$M_r = \underbrace{E \Sigma}_{E_r} D (E \Sigma)^{-1} = \underbrace{E \Sigma}_{E_r} D \underbrace{\Sigma^{-1} E^T}_{E_r^{-1}}$$

$$M_r = \underbrace{E \Sigma}_{E_r} D \underbrace{\Sigma^{-1} E^T}_{E_r^{-1}} = E (\Sigma D \Sigma^{-1}) E^T = \boxed{E D_r E^T}$$

$$\hat{x} = M_T(x) f(x)$$

$$= \underbrace{(E \Sigma)}_{E_r} \underbrace{D (\Sigma^{-1} E^T)}_{E_r^{-1}} f(x)$$

$$E_r = E \Sigma$$

$$\Sigma = \begin{bmatrix} n^T r & 0 \\ e^T r & 1 \end{bmatrix}$$

$$\Sigma^{-1} = \begin{bmatrix} \frac{1}{n^T r} & 0 \\ \frac{e^T r}{n^T r} & 1 \end{bmatrix}$$

$$= E (\underline{\underline{\Sigma D \Sigma^{-1}}}) E^T f(x)$$

$$[\lambda_n = \lambda_e]$$

$$\Sigma D \Sigma^{-1} = \begin{bmatrix} n^T r & 0 \\ e^T r & 1 \end{bmatrix} \begin{bmatrix} \lambda_n & 0 \\ 0 & \lambda_e \end{bmatrix} \begin{bmatrix} \frac{1}{n^T r} & 0 \\ -\frac{e^T r}{n^T r} & 1 \end{bmatrix}$$

$$\Sigma D \Sigma^{-1} = \begin{bmatrix} \lambda_n & 0 \\ \frac{e^T r (\lambda_n - \lambda_e)}{n^T r} & \lambda_e \end{bmatrix}$$

$$\lambda_n = 1 - \frac{1}{\lambda}$$

$$\lambda_e = 1 + \frac{1}{\lambda}$$

$$\underline{\lambda_n - \lambda_e = 2}$$

$$= \begin{bmatrix} n & e \end{bmatrix} \begin{bmatrix} \lambda_n & 0 \\ -\frac{2 e^T r}{r(x) n^T r} & \lambda_e \end{bmatrix} \begin{bmatrix} n^T \\ e^T \end{bmatrix} f(x)$$

$$= \begin{bmatrix} n & e \end{bmatrix} \left[\lambda_n n^T f(x) + \left(-\frac{2}{r(x)} \frac{e^T r}{n^T r} n^T f(x) \right) + \lambda_e e^T f(x) \right]$$



$$\dot{\hat{x}} = (n \ e) \begin{bmatrix} \lambda_n n^T f(x) \\ -\frac{2}{r(x)} \cdot \frac{e^T r}{n^T r} n^T f(x) + \lambda_e e^T f(x) \end{bmatrix}$$

$$= \underline{n [\lambda_n n^T f(x)]} + \left[e (\lambda_e e^T f(x)) - e \left(\frac{2}{r(x)} \cdot \frac{e^T r}{n^T r} n^T f(x) \right) \right] \stackrel{0}{=} 0 \quad ?$$

$$r(x) = \underline{1}, \quad \underline{e^T f(x) = 0} \quad \text{or} \quad \underline{n^T f(x) = \pm 1}$$

(collinearity issue)

$$\left[e (\lambda_e e^T f(x)) - e \left(\frac{2}{r(x)} \cdot \frac{e^T r}{n^T r} n^T f(x) \right) \right] \stackrel{r(x)=1}{=} 0 \quad ?$$

$$= e \ 2 \ e^T f(x) - e \ 2 \ \frac{e^T r}{n^T r} \cdot n^T f(x)$$

$$= 2e \left[e^T f(x) - \frac{e^T r}{n^T r} \cdot n^T f(x) \right] \stackrel{0}{=} 0 \quad ?$$

$$f(x) = c \cdot r(x)$$

$$= 2e \left[e^T (c \cdot r(x)) - \frac{e^T r}{n^T r} \cdot n^T (c \cdot r(x)) \right]$$

$$= \underline{2c} \ e \left[e^T r(x) - \frac{e^T r(x)}{n^T r(x)} \cdot n^T r(x) \right] = \underline{0}$$

Exercice #9

$\hat{x}_n, \lambda_e$

$$\hat{x} = \underbrace{\chi(x)}_{\substack{\uparrow \\ \text{max } \|u(x)f(x)\| = 2 \|f(x)\|}} M(x) f(x)$$

$$\hat{x}_n = \frac{M(x)f(x)}{\|M(x)f(x)\|}$$

$$\hat{x} = \|f(x)\| \hat{x}_n$$